

SMART AI IRRIGATION SYSTEM

ABSTRACT

This project is a Smart Agriculture Irrigation System that uses IoT sensors and Artificial Intelligence to automate watering and assist farmers in intelligent crop planning. The system collects real-time soil moisture and humidity data using a NodeMCU device placed in the field. The device sends this data to a backend server. The backend stores the data and sends structured crop and environmental information to an AI engine (Gemini API) for analysis. Based on crop type, soil condition, crop age, and sensor readings, the AI decides whether irrigation is required. The backend then sends a motor ON or OFF command back to the device. In addition to automation, the system stores historical crop data. This allows AI-based crop rotation suggestions and future planning recommendations to improve soil health and maximize yield. The system supports multiple fields but uses a single IoT device. Users can switch the active field from the backend interface. Safety logic ensures that irrigation will not start unless a field is selected.

SYSTEM FLOW (STEP BY STEP)

- **Field Creation:** Crop details such as crop name, soil type, and seed date are stored in the backend.
- **Active Field Selection:** The user selects which field is currently connected to the IoT device.
- **Sensor Data Collection:** NodeMCU reads soil moisture and humidity and sends data to the backend.
- **Backend Processing:** The backend validates data, stores logs, and prepares structured input for AI.
- **AI Decision Making:** AI analyzes crop stage and environmental data and decides irrigation requirement.
- **Motor Control:** Backend sends motor ON/OFF command to the NodeMCU.
- **Daily Analysis:** The system summarizes daily data and sends it to AI for advanced recommendations.
- **Historical Storage:** Crop cycles are stored for future crop planning and soil management.